

Paper 1

1. Describe anatomy of caudal epidural space. Describe the technique, advantages and complications of caudal epidural anesthesia in a 2 year old child.
2. Describe physiological changes during pregnancy what is the etiology and pathogenesis of pregnancy induced hypertension.
3. Describe in brief the Physiology and importance of intra cranial pressure. What is the clinical presentation and management of raised ICP.
4. What is oxygen therapy? What are the various levels/modes of oxygen therapy and methods being used for its judicious use?
5. An elderly overweight patient, who had undergone laparoscopic cholecystectomy under general anesthesia with isoflurane and vecuronium and had also received morphine sulphate (6 mg). She was reversed with empirical dose of Cholinesterase inhibitor, although breathing normally on arrival in RR, later developed respiratory failure. What could be the cause? Is there still any residual blockade and further how the case could be managed?
6. What are the basic ventilatory modes? Write about the management of mechanical ventilation under following heads : Initiation /Monitoring /Various strategies to improve the outcome /Weaning
7. Draw a neat labelled diagram of NMJ. Discuss the physiology of neuromuscular transmission and briefly describe the mechanism of action of non-depolarising and depolarising muscle relaxants
8. Describe the Physiology of haemostasis. How do you evaluate hemorrhagic diathesis? What precautions will you take before surgery?
9. Discuss the electrical hazards in the operation room and the various steps to prevent and minimize them.
10. Write precisely about the pain pathway during labour. Enumerate various methods of painless labour. Describe in detail lumbar epidural analgesia along with the recent trends.
11. Describe the anatomy of trachea-bronchial tree. Discuss the anesthetic management of a 35 years adult scheduled for thoracotomy.
12. Write a note on the evolution of anesthesia machine from Boyle's apparatus to modern anesthesia workstation. Enumerate various safety features of anesthesia machine and write in detail about anti hypoxic devices.
13. Describe anatomy of brachial plexus. Enumerate different approaches of brachial plexus block. Discuss its indications, techniques and complications.
14. Describe physiological changes of pregnancy. What is etiology pathogenesis and pathological alteration in pregnancy induced hypertension.
15. Describe autonomic nervous system with its clinical significance.
16. Enumerate physiological functions of the kidney. Describe nephron with the help of labelled diagram and Physiology of GFR and discuss the nephrotoxic factors and insults in peri operative.
17. Describe anatomy of IJV and write method of cannulation
18. Describe anatomy and physiology of NMJ junction and neuromuscular monitoring

Paper 2

1. Discuss management of peri operative pulmonary Oedema.
2. Describe anaesthetic management of a case of intracranial space occupying lesion
3. Describe preoperative evaluation and anaesthetic management of a patient of diabetes mellitus scheduled for emergency exploratory laparotomy.
4. What are the anesthetic challenges of neuro anesthesia. Describe the peri operative preparation and anesthetic management of a patient of S.O.L with raised icp for craniotomy.
5. Describe anatomical and physiological changes of pregnancy. discuss that as the management of a 39 weeks pregnant lady with positive Covid report for LSCS
6. Discuss preoperative preparations and anaesthetic arrangement of 55 years old patient with hypertension and coronary stent on beta blocker and dual anti platelet drugs to be taken for laparoscopic cholecystectomy.
7. Describe Peri operative evaluation of a patient with diabetes mellitus scheduled for major abdominal surgery. briefly describe the methods of maintaining in the operator blood sugar levels
8. Discuss anesthetic management of a patient with drug induced impairment of liver function posted for LSCS.
9. Discuss peri operative management of neonate with congenital diaphragmatic hernia.
10. Describe renal function tests. Discuss anesthetic management of a patient scheduled for kidney transplant surgery.
11. Discuss the anesthetic challenges in a 55 years patient with history of hypertension and Ischaemic heart disease with coronary stents on b-blockers, dual antiplatelet drugs and statin.
12. Describe the importance of pulmonary function tests in pre anesthetic assessment of a patient. Describe various post operative pulmonary complications.
13. Discuss anesthetic considerations and perioperative management of a patient .
14. Discuss anesthetic considerations in morbidly obese patient posted for laparoscopy cholecystectomy
15. Discuss peri operative management of neonatal with congenital diaphragmatic hernia
16. Discuss problems and anaesthesia management of neonate with congenital diaphragmatic hernia
17. Discuss problem and anesthesia management of 75 year old man with history of diabetes mellitus and hypertension scheduled for TURP surgery
18. Discuss problems, preparation and anesthesia management of a patient scheduled for pheochromocytoma surgery

Paper 3

- 1) Discuss the systemic changes which occur in old age. Describe anesthetic management of a patient with prostatic enlargement to be taken for TURP
- 2) Discuss preoperative assessment and anesthetic management of a patient of obstructive jaundice posted for emergency abdominal surgery
- 3) Discuss anesthetic management and post operative complications of tonsillectomy in a 7 year old child
- 4) Define diabetes mellitus. Classify it. What is importance of HBA1C. Describe the management of diabetic ketoacidosis.
- 5) What are the anatomical and physiological changes of aging. Describe peri operative care of frail elderly.
- 6) How a newborn differs from adult. Discuss the anaesthetic management of a newborn baby of Covid positive mother for trachea esophageal fistula repair.
- 7) A 25 year old gravida 3, para 2 is admitted in her 34th week of pregnancy with painless vagina bleeding hypotension and tachycardia discuss the peri operative management of this patient posted for emergency LSCS.
- 8) Describe the management of a 3 year old child with foreign body aspiration posted for bronchoscopy
- 9) Discuss the peri operative preparation and anaesthetic management of a child posted for cleft palate repair
- 10) Write precisely the Physiology of CSF circulation. Describe the anesthetic challenges and management of a 6 month baby of hydrocephalus scheduled for VP shunt
- 11) Write about anesthetic considerations in geriatric patients. Discuss the anesthetic management of 80 years old patient with COPD and coronary artery disease with low ejection fraction scheduled for inguinal hernia surgery
- 12) Describe the anesthetic challenges in morbidly obese patients. Discuss the anesthetic management of a 180 kg patient scheduled for bariatric surgery
- 13) Discuss anesthetic considerations in a 1 year old child with vegetable foreign body in right bronchus
- 14) Discuss anesthetic considerations and intra-operative management for a patient of chronic renal failure posted for emergency laparotomy
- 15) What are the basic principles of management of a neurosurgical patient? Discuss the common post operative complications after craniotomy
- 16) Describe preoperative evaluation preparation and peri operative management of hyperthyroid patient posted for subtotal thyroidectomy
- 17) Describe preoperative preparation evaluation and anaesthetic management of the newborn having congenital diaphragmatic hernia posted for repair
- 18) Describe preoperative preparation evaluation and anesthesia management of a 80 year old hypertensive and diabetic patient posted for TURP

Paper 4

- 1) Describe role of epidural for labour pain. Discuss various anesthesia techniques for caesarean delivery in patients with epidural catheter in place for labour analgesia
- 2) Discuss anesthetic management and peri operative infection control in covid 19 positive patients scheduled for emergency surgery.
- 3) Discuss the recent concept of sepsis and its management
- 4) What are Neuraxial blocks? Write about their clinical considerations and complications.
- 5) What is ERAS? What are the peri operative measures taken to achieve it
- 6) What is COVID-19? Write about its pathophysiology and also describe a treatment plan and management of COVID-19 patients.
- 7) Discuss the management of a patient with ventricular fibrillation. Discuss the changes in the latest AHA guidelines from the previous AHA guidelines.
- 8) Describe the clinical presentation management and anaesthetic management of a patient of diabetic ketoacidosis posted for surgical debridement.
- 9) Briefly discuss the latest recommendation on intensive care management of coronavirus disease 2019.
- 10) Write precisely liver function tests. Discuss the anesthetic management of liver transplantation.
- 11) Describe the anesthetic considerations in a patient with severe mitral stenosis. Discuss the anesthetic management including preoperative preparation and post operative care of a patient with mitral valve replacement (5 years back) on oral anticoagulants scheduled for hip surgery.
- 12) Write an overview of pain clinic. Discuss the various modalities available for management of low back pain
- 13) Define ARDS (acute respiratory distress syndrome). What is the differential diagnosis? Discuss evidence based strategy for ventilatory management of a patient of ARDS
- 14) Describe the pathophysiology underlying post operative pain. Discuss various modalities for inhibition of peripheral and central sensitization.
- 15) Discuss the recent concept of sepsis and its management
- 16) describe the pathophysiology of adult respiratory distress syndrome. Describe its management with special emphasis on evidence based guidelines.
- 17) Discuss the various modalities of post operative pain management. What is the peri operative analysis strategy for management of post thoracotomy pain?
- 18) Discuss the principles of management of a patient with severe traumatic injury with specific reference to the importance of golden hour and rapid intervention.